Probit Regression Function

Applications in Government & Politics

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Government/Politics Dataset | Implementation

Research question: to what extent do socioeconomic factors, specifically age, education, and income, predict voter turnout in American national elections?

We used a dataset provided by the American National Election Studies, detailing whether individuals voted or not in the 2024 election. The dataset had over 44 variables it considered, such as occupation and religion.

Clean Data Explanation

Translating information from the codebook, we considered the variables below.

Table 1: Variable Definitions

Variable	Name	Measurement	Coding / Description
VCF0702	Turnout	Binary	1 = Voted, 0 = Did not vote (recode)
VCF0101	Age	Continuous	Respondent age in years
VCF0110	Education	Categorical (Ordinal)	1 = LessHS, 2 = HighSchool, 3 = SomeCollege, 4 = CollegePlus
VCF0114	Income	Ordinal (treated numeric)	Higher values = higher income group
VCF0301	Party ID	Categorical (7-point scale)	StrongDem → StrongRep (7 categories)
VCF0310	Political Interest	Ordinal (1–3)	1 = Low, 2 = Medium, 3 = High interest
VCF0009z	Survey Weight	Continuous	ANES full-sample weight (post-stratification)

VCF0009z, considers if a survey accidentally talks to too many men and not enough women, the women's answers are given a higher weight (they count for more than one person) and the men's answers are given a lower weight (they count for less than one person).

We cleaned the dataset to only use the variables above

Topic 4 | Implementation

Exploratory Data Analysis

```r
chisq.test(table(df_clean$vote, df_clean$education))
```

```
	Pearson's Chi-squared test

data:  table(df_clean$vote, df_clean$education)
X-squared = 3910.4, df = 3, p-value < 2.2e-16
```

```r
chisq.test(table(df_clean$vote, df_clean$party_id))
```

```
	Pearson's Chi-squared test

data:  table(df_clean$vote, df_clean$party_id)
X-squared = 3694.5, df = 6, p-value < 2.2e-16
```

- For education, the test statistic is very large $X^2 = 3910.4$ with a p-value effectively equal to zero. We can confidently reject the idea that voting behavior is independent of education level.
- For party identification, the chi-squared test $X^2 = 3694.5$ also produces a near-zero p-value, indicating a strong relationship between party ID and turnout.

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```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!


Call:
glm(formula = vote ~ age + education + income + party_id + interest,
    family = binomial(link = "probit"), data = df_clean, weights = weight)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)          -1.2271752  0.0323903  -37.89   <2e-16 ***
age                   0.0145019  0.0003513   41.28   <2e-16 ***
educationSomeCollege -0.2651494  0.0191872  -13.82   <2e-16 ***
educationHighSchool  -0.5935501  0.0171903  -34.53   <2e-16 ***
educationLessHS      -0.8753739  0.0229323  -38.17   <2e-16 ***
income                0.1417566  0.0047471   29.86   <2e-16 ***
party_idIndLeanDem    0.3025451  0.0226679   13.35   <2e-16 ***
party_idIndLeanRep    0.3480854  0.0235716   14.77   <2e-16 ***
party_idWeakDem       0.3703462  0.0201312   18.40   <2e-16 ***
party_idWeakRep       0.4335910  0.0221265   19.60   <2e-16 ***
party_idStrongDem     0.6847337  0.0212788   32.18   <2e-16 ***
party_idStrongRep     0.7786772  0.0244692   31.82   <2e-16 ***
interest              0.4008260  0.0075930   52.79   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 74508  on 61812  degrees of freedom
Residual deviance: 61211  on 61800  degrees of freedom
AIC: 60653

Number of Fisher Scoring iterations: 4
```

Latent scores.

Research Conclusion

```r
mfx <- margins(model1)
summary(mfx)
```

factor	AME	SE	z	p	lower
age	0.0039	0.0001	43.1511	0.0000	0.0037
educationHighSchool	-0.1548	0.0041	-37.3483	0.0000	-0.1629
educationLessHS	-0.2449	0.0065	-37.5029	0.0000	-0.2577
educationSomeCollege	-0.0621	0.0044	-13.9750	0.0000	-0.0708
income	0.0381	0.0013	30.1869	0.0000	0.0356
interest	0.1076	0.0019	56.2491	0.0000	0.1039
party_idIndLeanDem	0.0947	0.0071	13.3597	0.0000	0.0808
party_idIndLeanRep	0.1080	0.0073	14.8308	0.0000	0.0938
party_idStrongDem	0.1975	0.0063	31.4816	0.0000	0.1852
party_idStrongRep	0.2192	0.0068	32.4330	0.0000	0.2059
party_idWeakDem	0.1145	0.0063	18.1659	0.0000	0.1021
party_idWeakRep	0.1324	0.0068	19.5564	0.0000	0.1191

Research Conclusion

The average marginal effect (AME) of 0.0039 means that each additional year of age increases the probability of voting by about **0.4 percentage points**, holding other factors constant. This suggests that older individuals are more likely to vote.

Compared to individuals with a college degree (the baseline group), those with lower levels of education are significantly less likely to vote. For example, having only a high school education decreases the probability of voting by about **15.5 percentage points**

The AME of 0.0381 shows that moving up one income category increases the probability of voting by about **3.8 percentage points**.

Compared to independents, individuals who identify with a political party, especially strong partisans, are substantially more likely to vote. For example, strong Democrats and strong Republicans are about **20–22 percentage points more likely** to vote than independents.

Similarly, **political interest** has one of the largest effects in the model. The average marginal effect of 0.1076 indicates that moving up one level of interest increases the probability of voting by about **10.8 percentage points**.

Overall, while socioeconomic resources are important, these results show that political engagement and partisan identity are equally powerful predictors of turnout.